2nd International Symposium on the Recent Developments in Plate Steels, June 3-6, 2018, Orlando, FL

Rolling pressures during the Hot-Roll of three different steel plates: Experimental, Theoretical and FEA

M.F. Buchely[1], X. Wang[2], D.C. Van Aken[1], R.J. O'Malley[1], S. Lekakh[1], K. Chandrashekhara[2]

[1] Department of Materials Science and Engineering.
Missouri University of Science and Technology,
Rolla, MO 65409, USA

[2] Department of Mechanical and Aerospace Engineering.
Missouri University of Science and Technology,
Rolla, MO 65409, USA

Keywords: Hot-rolling, Instrumented Rolling Mill, Rolling pressure, Steel plates

NOTATION

α	Angle of contact at the plane of entry		L_p	Arc of contact ($\approx \sqrt{R\Delta h}$)
ε_m	Mean strain during rolling		L_p'	Corrected arc of contact ($\approx \sqrt{R'\Delta h}$)
ε_{pl}	Plastic strain		h_1	Entry thickness
$\dot{\varepsilon}_m$	Mean strain rate during rolling		h_2	Exit thickness
$\dot{\varepsilon}_o$	Strain rate of reference in Johnson-cook model		Δh	Draft ($= h_1 - h_2$)
$\dot{\varepsilon}_{pl}$	Plastic strain rate		h_m	Mean thickness ($= (h_1 + h_2)/2$)
ϕ	Angular co-ordinate of the arc of contact		h_n	Thickness at the neutral plane
ϕ_n	Angle of the neutral point		M	Torque during rolling
λ	Angle of friction ($\mu = \tan\lambda$)		m	Johnson-Cook parameter
μ	Coefficient of friction		n	Johnson-Cook parameter
σ	Flow stress in plane strain		p_r	Normal roll pressure
σ^*	Uniaxial flow stress		R	Roll radius
$\bar{\sigma}$	Mean flow stress during rolling		R'	Radius of curvature of elastically deformed rolls
σ_x	Stress in the direction of rolling		T	Temperature
σ_y	Stress normal to the rolling direction		T_m	Melting temperature
τ	Frictional stress		T_r	Temperature of reference
A	Johnson-Cook parameter		v_r	Peripheral rolling speed
a	Lever arm for torque generation		w_1	Initial width
B	Johnson-Cook parameter		w_2	Final width
C	Johnson-Cook parameter		w_m	Mean width ($= (w_1 + w_2)/2$)
F	Roll force			

INTRODUCTION

Calculation of rolling pressure is an important subject in hot rolling of plate, and this information is useful for rolling mill operation and design[1]. Many theories related to the calculation of rolling pressure are available in the literature. For example, Table I shows a summary of many theoretical, analytical and empirical expressions that have been used for hot rolling purposes. In this summary, intricate theories[2,3] have not been considered because they involved solution of complex differential equations, which in many cases makes them impractical for industrial processes.

Table I. Summary of theoretical, analytical and empirical expressions to calculate rolling pressure (p_r) and torque (M) during hot rolling.

Index	Name	Equation
A	Simple slipping friction situation [4]	$p_r = \frac{\bar{\sigma}}{Q}\left[e^{Q}-1\right]$, and $M = 2\,p_r w_m L_p' a$; where a is the lever arm $\approx L_p'/2$ and $Q = \mu L_p'/h_m$
B	Simple sticking friction situation [5]	$p_r = \bar{\sigma}\left[\frac{L_p'}{4h_m}+1\right]$
C	Ekelund [6]	$p_r = \bar{\sigma}\left[1+\frac{1.6\mu L_p' - 1.2\Delta h}{h_1+h_2}\right]$
D	Sims [7]	$p_r = \bar{\sigma}Q_p$, and $M = 2RR'\sigma Q_g$; where Q_p and Q_g are graphically determined
E	Green-Wallace [8,9]	$p_r = \frac{\bar{\sigma}}{2}\left[\frac{L_p'}{h_m}+\frac{h_m}{L_p'}\right]$
F	Ford-Alexander [9,10]	$p_r = \frac{\bar{\sigma}}{2}\left[\frac{\pi}{2}+\frac{L_p'}{h_1+h_2}\right]$, and $M = \frac{w_m L_p'^2 \sigma}{2}\left[\frac{\pi}{2}+\frac{2L_p'(h_1+2h_2)}{3(h_1+h_2)^2}\right]$
G	Tselikov [11,12]	$p_r = \bar{\sigma}\frac{2h_2}{\Delta h(Q-1)}\left(\frac{h_n}{h_2}\right)\left[\left(\frac{h_n}{h_2}\right)^{Q}-1\right]$, where $Q = \mu\sqrt{R/\Delta h}$ and $\frac{h_n}{h_2} = \left[\frac{1+\sqrt{1+(Q^2-1)(h_1/h_2)^{Q}}}{Q+1}\right]^{1/Q}$
H	Golovin-Tiagunov [12]	$p_r = \bar{\sigma}\left[1+\mu\left(\frac{L_p'}{h_m}-1\right)\right]$
I	Orowan-Pascoe [12,13]	$p_r = \bar{\sigma}\left[\frac{\pi}{4}+\frac{cL_p'}{2h_2}-\frac{h_m h_2}{3cw_m L_p'}\left(\frac{cL_p'}{h_2}-0.2\right)^3\right]$, where c is graphically determined
J	Denton-Crane [4,14]	$p_r = \frac{\bar{\sigma}}{2}\left[0.655+0.265\frac{L_p'}{\sqrt{h_1 h_2}}\right]$, and $M = \frac{w_m L_p'\sigma}{2}\left[1.59+0.44\frac{L_p'}{\sqrt{h_1 h_2}}\right]$
K	Geleji [12]	$p_r = \bar{\sigma}\left[1+c\mu\frac{L_p'}{h_m}\sqrt[4]{v_r}\right]$, where c is graphically determined
L	Roberts [15]	$p_r = \bar{\sigma}\exp\left(\mu\frac{L_p'}{2h_m}\right)$
M	Schey [16]	$p_r = \bar{\sigma}Q_p$, where Q_p graphically determined

In order to solve any equation in Table I, it is important to know the behavior of the material at different rolling conditions, especially at different percentage of reduction (strain), velocities of deformation (strain-rates) and temperatures [5,12]. For this reason, many researches have complemented their theoretical analysis with complex nomograms, tables and/or curves, to relate a mechanical property (particularly uniaxial stress) at different rolling conditions. However, those curves are only valid for a specific material, and most of the time, extrapolation of the data predicts wrong rolling settings (e.g. rolling pressures).

On the other hand, complete models of rolling processes using Finite-Element simulations have gained recently attention because they can model more complex geometries and more realistic contact analysis[17]. Still, 3D FE models could take hours to obtain a reliable solution (depending of mesh and complexity). Additionally, good constitutive models must to be applied to the material to obtain comparable data. One constitutive equation commonly used in hot-working simulations is the Johnson-Cook strength model [18,19].

Between analytical equations and FE simulations, simplified 1D models represents another approaches to predict rolling parameters for simple processes (flat or round rolling) [2,20].

In this paper, a comparative analysis of rolling pressures during hot rolling was performed using three different approaches: 1D analytical model, empirical equations, and FE simulations. For each approach, same material model and friction conditions were assumed. Therefore, Johnson-Cook strength model and Coulomb's friction were selected. For this comparative analysis, three different steels were studied: AISI/SAE 15V38, ASTM A992, and AISI/SAE 1018. Hot rolling tests were performed using a STANAG Instrumented Rolling mill, set up in the 2-high roll configuration, and using 5 in diameter, tool steel rolls. This equipment was instrumented with load cells, torque cells and IR thermometers to obtain valuable data during hot rolling. Rolling reductions were approximately 42% in thickness at 32 RPM and at a starting temperature of 1050ºC. Experimental data was processed and compared to predictions among different approaches.

Table II. Chemical composition of studied steels (%wt, %Fe balance).

	C	Si	Mn	P	S	Cr	Mo	Ni	Cu	Al	Nb	V
15V38	0.4	0.53	1.38	0.01	0.06	0.13	0.02	0.08	0.16	0.02	0.002	0.09
A992	0.09	0.31	1.04	0.02	0.02	0.43	0.05	0.32	0.28	-	0.08	0.04
1018	0.16	0.11	0.77	0.01	0.04	0.04	-	-	0.27	>0.01	>0.01	-

MATERIALS AND METHODS

Hot-rolling process

Three materials were selected to this hot-rolling study: AISI/SAE 15V38, ASTM A992, and AISI/SAE 1018 steel. First steel (15V38) was extracted from an as-cast round billet, 9in diameter. Second steel (A992) was extracted from a flange of an as-cast beam blank. Last steel (1018) was chosen as reference, and was extracted from commercial cold-rolled rectangular bars, 2in wide x 0.5in thick. Chemical composition of these steels is shown in Table II.

For the hot rolling test, material was sectioned in 5in long, 2in wide, 0.5in thick rectangular coupons. A 60deg angle was machined in one of the coupon edges, to help the first bite of rolling, as shown in Figure 1. Hot rolling was performed using a STANAG TA-315 Instrumented Rolling mill, set up in the 2-high roll configuration, and using 5in diameter, tool steel rolls. This equipment was instrumented with two load cells (up to 65,000lbf), two torque cells (up to 30,000in.lbf) and entry and exit IR thermometers (380ºC to 1600ºC) to obtain valuable data during hot rolling. Additionally, the angular velocity of the rolls and load of the motor are measured during the process. Figure 1 shows a schematic view of the rolling system.

Coupons were austenitized at 1100ºC per 1hr. After that, they were rolled, reducing approx. 42% of thickness in one pass. Rolling data was saved for each coupon, as shown in Figure 2, and post-processed using the software MATLAB. Torque signals were added, whereas other pair of signals were averaged (force and temperature).

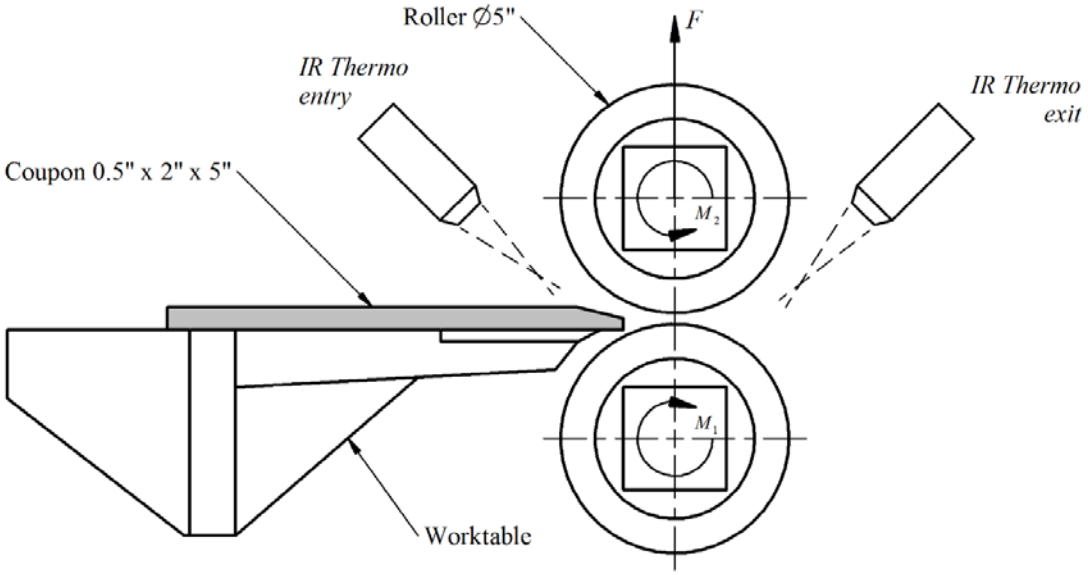
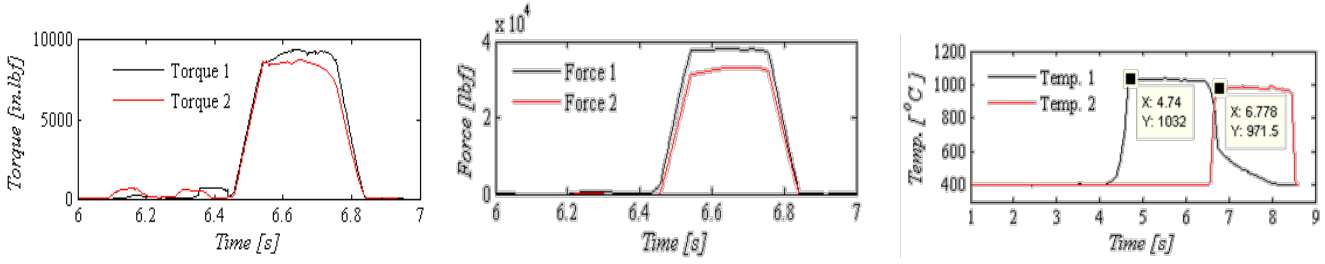


Figure 1. Schematic view of the rolling system. Instrumentation consists of load cells, torque cells and IR thermometers.



(a) (b) (c)

Figure 2. Example of data captured during hot-rolling of 15V38 (coupon 4) steel: (a) torque signals, (b) force signals, and (c) temperatures.

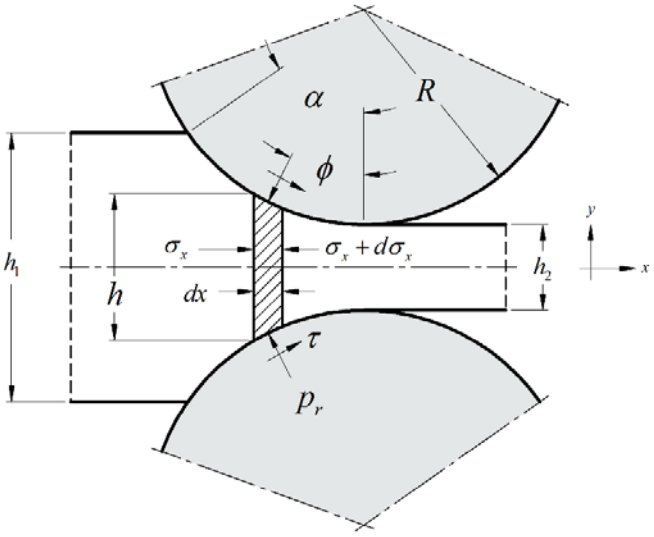


Figure 3. Model of rolling in plane strain

1D analytical model

Figure 3 shows a simplified 1D model of hot rolling, assuming plane strain conditions. All nomenclature used in this figure and in the following equations are presented in the Notation table at the beginning of this paper. Based on this model, the differential equation of equilibrium during plane strain rolling is given by [1]:

$$\frac{d\left(\sigma_x h\right)}{dx} = -p_r \frac{dh}{dx} \pm 2\tau \; ; \tag{1}$$

or, in terms of the angular co-ordinate ϕ [2]:

$$\frac{d\left(\sigma_x h\right)}{d\phi} = 2Rp_r \sin\phi \pm 2R\tau\cos\phi \; ; \tag{2}$$

where the $\pm$ sign indicates that the equation above describes the condition of equilibrium between the neutral point and the exit (using the positive sign), as well as between the neutral point and the entry (using the negative sign).
Assuming Huber-Mises condition for plasticity, as follows

$$\sigma_y - \sigma_x = \frac{2}{\sqrt{3}}\sigma^* = \sigma \; ; \tag{3}$$

where principal stress in *y* direction σ_y is given by:

$$\sigma_y = p_r \mp \tau\tan\phi \; ; \tag{4}$$

and using slipping condition of friction, given by Coulomb's law, as follows:

$$\tau = \mu p_r \; ; \tag{5}$$

differential equation (2) is transformed to:

$$\frac{d\left(\sigma_x h\right)}{d\phi} = 2R\left(\sigma_x + \sigma\right)\cos\phi\tan\left(\phi \pm \lambda\right). \tag{6}$$

In order to solve equation (6), the material flow stress (σ^* or σ) must be known. In this case, Johnson-Cook strength model[18] was used because this model combines uniaxial flow stress, plastic strain, strain-rates and temperatures in the same equation, as follows:

$$\sigma^* = \frac{\sqrt{3}}{2}\sigma = \left(A + B\varepsilon_{pl}^n\right)\left(1 + C\ln\left(\frac{\dot{\varepsilon}_{pl}}{\dot{\varepsilon}_0}\right)\right)\left(1 - \left(\frac{T - T_r}{T_m - T_r}\right)^m\right) \tag{7}$$

Table III. Johnson-Cook parameters for the studied materials.

	A [MPa]	*B [MPa]*	*n*	*C*	*m*	$\dot{\varepsilon}_0$ *[s^{-1}]*	T_r *[^{o}C]*
15V38	16.9428	45.9149	0.5554	0.6224	0.6394	0.0299	1000
A992	10.6456	69.6988	0.4185	0.4139	0.5563	0.0551	1000
1018	8.9254	84.2933	0.4028	0.2827	0.7548	0.0291	1000

Table III shows Johnson-Cook parameters of the studied steels, which were determined by the authors in previous experiments using tensile tests at high temperatures and medium strain-rates [21,22].

Assuming that elastic strains are negligible, plastic strain and plastic strain-rates can be approximated to the mean strain and mean strain-rates during rolling, as follows [1,5]:

$$\varepsilon_{pl} \approx \varepsilon_m = \ln\frac{h_1}{h_2}, \text{ and} \tag{8}$$

$$\dot{\varepsilon}_{pl} \approx \dot{\varepsilon}_m = v_r\sqrt{\frac{1}{R\Delta h}}\ln\frac{h_1}{h_2}. \tag{9}$$

To simplified the model's solution, a constant and mean flow stress ($\bar{\sigma}$) during rolling was assumed. This can be calculated by integrating equation (7) in the range of the strain during rolling, as follows:

$$\bar{\sigma} = \frac{1}{\varepsilon_m}\int_0^{\varepsilon_m}\sigma d\varepsilon = \frac{2}{\sqrt{3}}\left(A+\frac{B\varepsilon_m^n}{1+n}\right)\left(1+C\ln\left(\frac{\dot{\varepsilon}_{pl}}{\dot{\varepsilon}_0}\right)\right)\left(1-\left(\frac{T-T_r}{T_m-T_r}\right)^m\right) \tag{10}$$

where, equation (8) was used in the integration.

Additionally, it was assumed a constant coefficient of friction during hot rolling of all materials. In preliminary rolling test at similar conditions of deformation, it was determined that $\mu \approx 0.29$ for the current rolling system.

System of equations (3) to (9) were programmed in MATLAB, and numerically solved using the Runge-Kutta 4 algorithm, with a uniform step size of 1E-6. It was assumed that stresses at the entry and exit side are zero, as boundary conditions. After solution, pressure during rolling (p_r) was determined as a function of ϕ .

Other theories and Empirical models

To compare results of the 1D model and experimental data, analytical and empirical expression by other authors were solved for the studied materials. Table I showed a summary of different expression for hot rolling analysis. All these equations were solved for the current rolling conditions. The mean flow stress ($\bar{\sigma}$) of each material was calculated using the Johnson-Cook based equation (10) and parameters in Table III at experimental strain, strain rates and temperatures. Additionally, for some expressions, same constant and experimental value of coefficient of friction was used ($\mu \approx 0.29$).

Finite-Element model

A Finite-element analysis of the rolling process was also modeled. In this case, a full-scale 3D model was processed using the explicit code ABAQUS, as seen in Figure 4. Rollers were modeled as rigid parts using 4-node rigid element (R3D4), and coupons were built using 8-node brick element (C3D8RT). For contact behavior, Coulomb's law with $\mu \approx 0.29$ was defined between surfaces. Viscoplastic behavior was modeled using the Johnson-Cook strength model (Equation (7) and Table III) for the different studied steels.

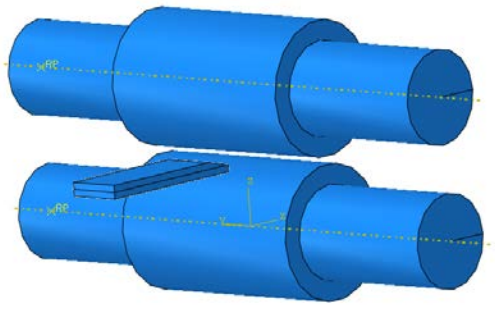


Figure 4. Finite-element model of the Rolling process programmed in the explicit code ABAQUS.

Table IV. Summary of rolling data of studied steels.

Material	Coupon	M_{exp} [*in.lbf*]	F_{exp} [*lbf*]	T [^oC]	h_1 [*mm*]	h_2 [*mm*]	w_1 [*mm*]	w_2 [*mm*]	$\dot{\varepsilon}_m$ [s^{-1}]
15V38	1	18001	35808	1029	12.54	7.21	51.17	53.29	6.40
15V38	2	17609	35197	1033	12.62	7.25	50.89	53.26	6.38
A992	3	15906	32655	1032	12.64	7.25	50.66	52.54	6.40
A992	4	16039	32800	1040	12.59	7.24	50.91	52.97	6.39
1018	5	14584	31664	1051	12.62	7.22	50.63	52.51	6.42
1018	6	14679	31371	1052	12.71	7.21	50.71	52.50	6.46

RESULT AND DISCUSSION

Table IV summarized experimental data rolling which was capturing during hot rolling process. For each steel, two identical coupons were rolled to verified consistency of data. As shown in Table IV, similar magnitude of torque and force were measured for same steel. Last column in Table IV shows mean strain rates during rolling, which were calculated using equation (9). Notice that similar levels of strain rates were applied to the different coupons during hot rolling.

In order to normalize the experimental force, rolling pressure during rolling were calculated, as follows:

$$p_{r_\exp} = F / \left(w_m L_p^{'} \right) \tag{11}$$

where w_m is the mean width, and $L_p^{'}$ is the corrected arc length based on the Hitchcock equation[20]:

$$R' = R\left(1 + \frac{2C_H F_{\exp}}{\Delta h \cdot w_m} \right) \tag{12}$$

where C_H is a constant depending of the type of rolls (C_H = 8.35E-8 in^2/lbf for steel rolls).

Table V shows the rolling pressures calculated by each coupon using the experimental data. In general, it can be seen that higher rolling pressures were necessary for forming 15V38 steel. A992 and 1018 steels seems to require same level of pressure. This tendency is related to the carbon content of these alloys.

Rolling pressures were also predicted after numerical solution of the 1D model. Figure 5 shows the predicted rolling pressure for the coupon 2 as a function of the contact angle. $\phi = 0$ represents the exit plane and $\phi = \alpha$ (approx. 3.33 DEG in this case) represents the entry plane. In the figure, the neutral point is also shown, which was found after solving equation (6) using the positive sign (for the exit plane) and the negative sign (for the entry plane) independently.

Using this solution, mean rolling pressure was calculated, as follows:

$$\overline{p}_{r_1\text{Dmodel}} = \frac{1}{\alpha}\int_0^{\alpha} p_r d\phi = \frac{1}{\phi_n}\int_0^{\phi_n} p_{r_exit} d\phi + \frac{1}{\alpha - \phi_n}\int_{\phi_n}^{\alpha} p_{r_entry} d\phi \tag{13}$$

Table V also shows the rolling pressures calculated for each coupon using the 1D model. Predictions seems to be in good agreement with experimental data.

Table V. Rolling pressures: experimental and predicted by 1D model

Coupon	$p_{r_\exp}$ [*psi*]	$\overline{p}_{r_1\text{D model}}$ [*psi*]
1	23876	24785
2	23464	24371
3	21922	21413
4	21959	21046
5	21254	20934
6	20864	20991

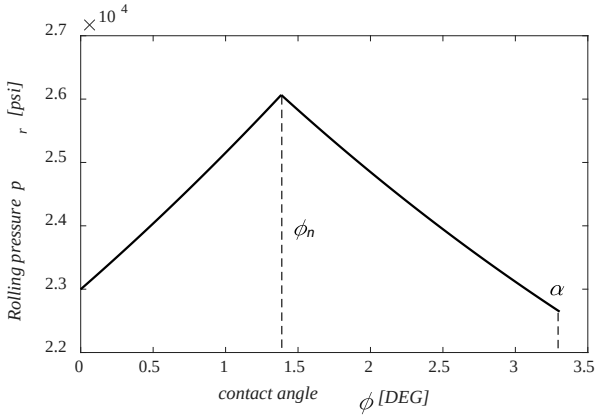


Figure 5. Predicted rolling pressure at different angles. Example for coupon 2.

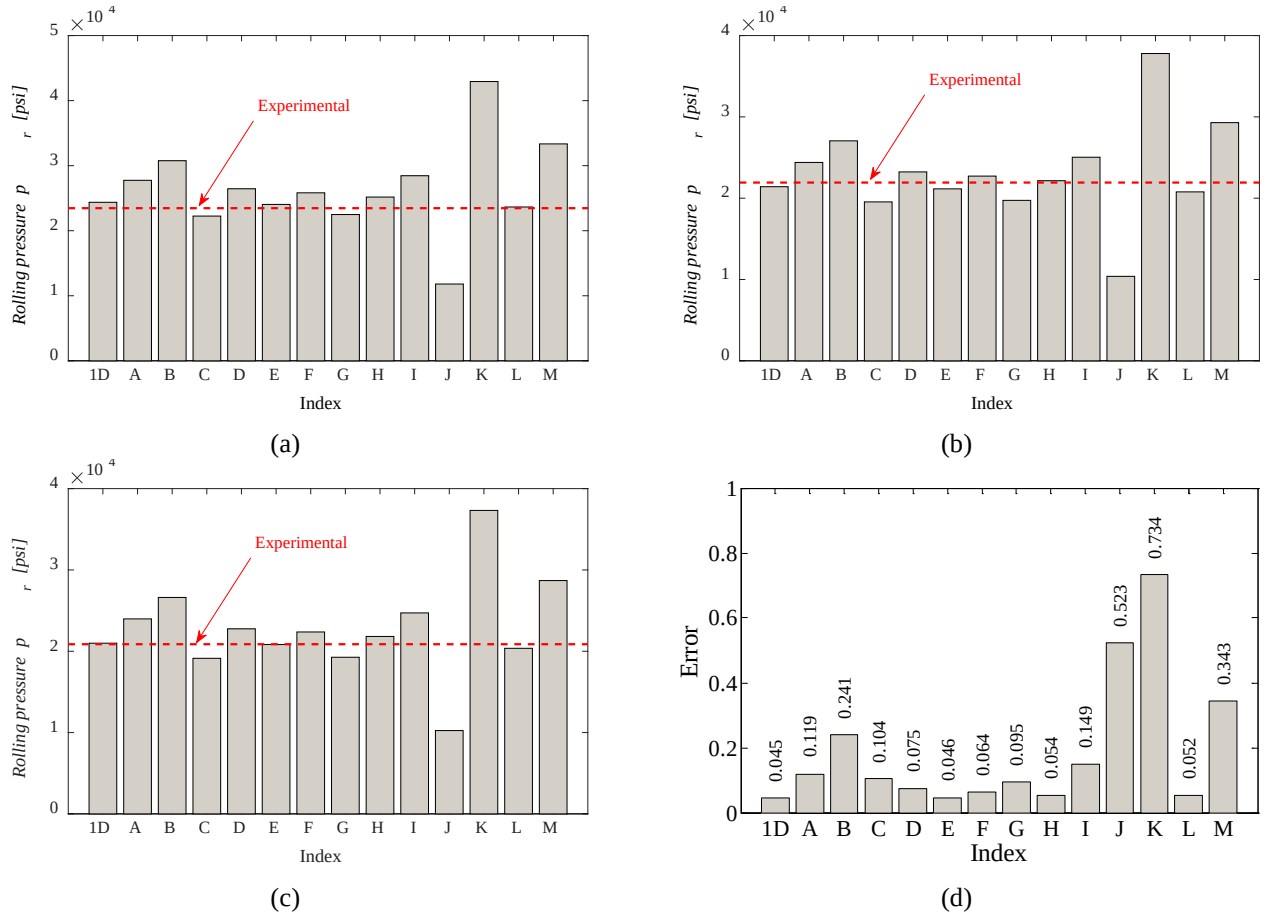


Figure 6. Comparison of the rolling pressure predicted among 1D model and theoretical approaches (see Index in Table 1): (a) 15V38 steel coupon 2, (b) A992 steel coupon 3, (c) 1018 steel coupon 6 and, (d) total error of each prediction.

Figure 6 shows rolling pressures predicted by other theoretical analysis (see Table I), and comparison among 1D model and experimental results. It can be seen that, for independent cases, some theoretical models could predict better results. For example, for the coupon 2 (Figure 6a), best prediction was obtained using Model L; for the coupon 3 (Figure 6b), best prediction is by Model H; and for coupon 6 (Figure 6c), best prediction is by Model E (see Table I for Index of each model).
To quantify the total error of each model prediction, the next simple error was calculated:

$$\text{Error} = \sum_{i=1}^{6} \frac{\left| p_{r_\exp} - p_{r_Index} \right|_i}{p_{r_\exp}\Big|_i} \tag{14}$$

where all predictions were compared and normalized by the experimental value, for each coupon *i*.
Figure 6d shows the calculated total error for each model. It can be seen that the global best predictions were obtained by the 1D model (4.5% error). It means that the 1D analytical model is a good representation of the physical rolling experiment for this simplified process. After this, the model E or Green-Wallace model (4.6% error) showed comparable and good predictions of the rolling pressures. It can be noticed that this model is based in geometrical assumptions, and it does not consider a coefficient of friction (see equation for Index E in Table I). Therefore, it represents an excellent choice for prediction the rolling pressures when the coefficient of friction is not known in the rolling process. After this, model L or Roberts model (5.2% error) and model H or Golovin-Tiagunov model (5.4% error) show reasonable predictions. Both models included coefficient of friction (Index L and Index H in Table I), and are also easy analytical expression that can be used for industrial purposes. On the other hand, two models seem to make poor predictions of rolling pressures. They are model K or Geleji model (73% error), and model J or Denton-Crane model (52% error). Both models used empirical information, which could have been calibrated for specific materials at specific rolling conditions (Index K and Index J in Table I). Additionally, former model involves the rolling velocity, which in this case was calculated based on the peripheral velocity of the rolls. However, those velocities are only equal at the neutral point, and it is known that exit velocity is higher that rolls velocity during rolling. Therefore, models which involves empirical constants seem to make bad predictions, and their use should be limited for different materials.

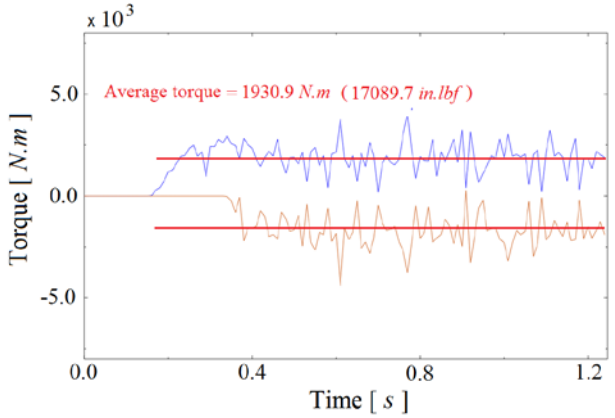


Figure 7. Predictions of torques in rollers by FE simulations for A992 steel, at the same experimental conditions.

Table VI. Comparison of torque predictions by FE simulations.

	M_{exp} [*in.lbf*]	M_{FE} [*in.lbf*]	Error
15V38	17805	19622	0.102
A992	15973	17090	0.069
1018	14631	16236	0.109

One model that it is usually known and technologically used for its simplicity and good predictions is the Sims model [1] or model D (Index D in Table I). In this study, Sims model shows an acceptable performance (7.5% error); however, it is higher than other predictions. This model consider sticking friction, and do not consider coefficient of friction for calculations [7], which has been shown to be a good assumption for hot rolling of big pieces (using bigger rollers) [1]. Although the high rolling temperatures, sticking friction could not have been reached in this work due to the use of small diameter rolls, and probably this is the reason why this model did not make better predictions.

Figure 7 shows an example of rolling torques in rollers predicted by the FE simulation for the A992 steel. Notice that signals have opposite sign due to different rotation of each roll. As can be seen, predicted torque signals shows a high level of noise, which is induced by the contact behavior and the mesh size of this FE model. However, an average value can be calculated for each roller, as shown in Figure 7. Table VI shows the torque comparison between experimental data and predicted by FE simulations. Notice that FE results overpredicted experimental data; however, errors in predictions were less than 10%, which are considerable acceptable for this work (comparing to predictions of theoretical models). FE predictions can be improved refining the mesh size of the rolling material, and/or modelling rollers as deformable bodies instead of rigid bodies. This analysis was not performed in the current paper, and it will be a topic for future work. Finally, Figure 8 shows comparison of geometries after rolling. Notice that FE simulations can predict spread and edges changes, which are not able to predict by simplified models. Therefore, although FE simulations did not predict the best results when torques were compared, they can be used for rolling products of different geometries and results shows more detail.

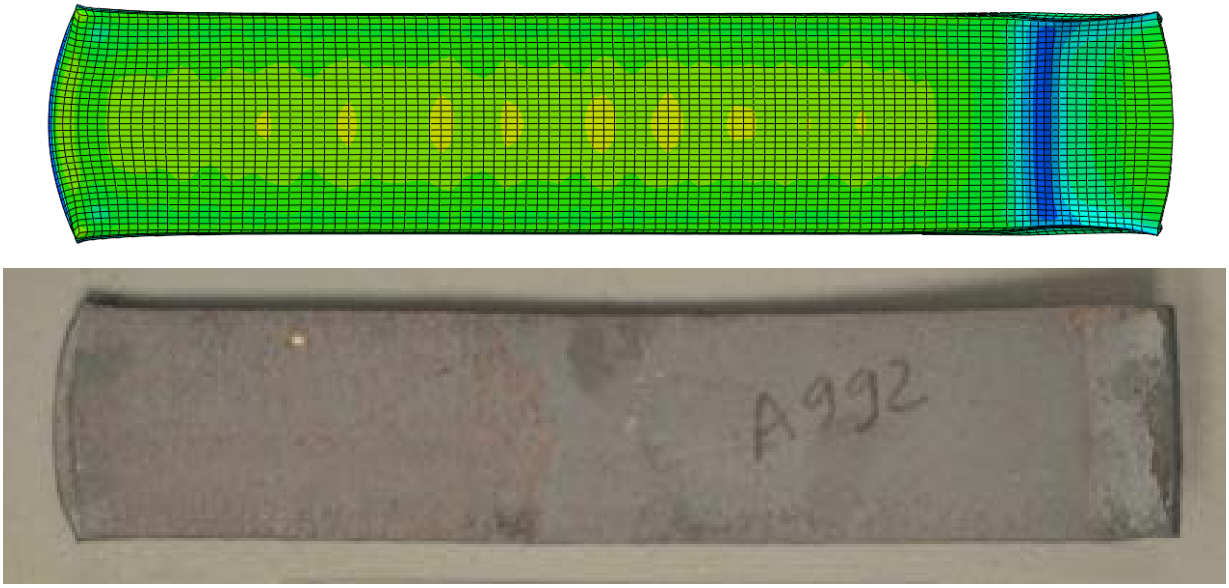


Figure 8. Comparison of final shape predicted by FE simulations and experimental coupon. A992 steel coupon 4.

CONCLUSIONS

In this paper, three different steels (AISI/SAE 15V38, ASTM A992, and AISI/SAE 1018) were hot rolled using an instrumented rolling mill. Valuable rolling information was reported for each steel, like force, torque, and rolling pressures.
Experimental rolling data was compared to predictions of three different approaches: 1D analytical model, empirical equations, and FE simulations using a common constitutive model governed by the Johnson-Cook strength model. Although, calibration methodology of the material model was not covered in this paper, it is very important to know the mechanical behavior of the material at different conditions of strain-rate and temperature.
1D model of the rolling process gave accurate predictions of rolling pressures, considering the simple flat rolling case presented in this work. If coefficient of friction is unknown, it is recommended to use Green-Wallace or Sims equations to predict rolling pressures during hot rolling. Also, if the coefficient of friction is known, Roberts or Golovin-Tiagunov equations seems to give acceptable predictions. If the aim is to know geometry changes and mass flows during rolling, it is recommended to use FE simulations to predict better features after processing.

ACKNOWLEDGEMENTS

This work is supported by the Peaslee Steel Manufacturing Research Center (PSMRC) at Missouri University of Science and Technology. The investigators acknowledge: Gerdau, Nucor, SSAB and US Steel for supporting Hot Working Modeling and Simulation project.

REFERENCES

1. Lenard, J. G. *Primer on flat rolling*. (2nd Ed., Elsevier Ltd, 2014).
2. Orowan, E. The calculation of Roll Pressure in Hot and Cold flat rolling. *Proc. Inst. Mech. Eng.* **150,** 140–167 (1943).
3. Malinowski, Z., Pietrzyk, M. & Lenard, J. G. Analysis of the flat-rolling process: one-dimensional and finite-element models. *J. Mater. Process. Technol.* **39,** 373–387 (1993).
4. Dieter, G. *Mechanical Metallurgy*. (McGraw-Hill, 1986).
5. Harris, J. N. *Mechanical working of metals*. (Pergamon Press, 1983).
6. Ekelund, S. Analysis of Factors Influencing Rolling Pressure and Power Consumption in the Hot Rolling of Steel. *Steel* **33,** 27–29 (1933).
7. Sims, R. B. The calculation of Roll Force and Torque in Hot Rolling Mill. *Proc. Inst. Mech. Eng.* **168,** 191–200 (1954).
8. Green, J. W. & Wallace, J. F. Estimation of Load and Torque in the Hot Rolling Process. *J. Mech. Eng. Sci.* **4,** 136–142 (1962).
9. Ginzburg, V. *Steel-rolling technology: theory and practice*. (Marcel Dekker, Inc, 1989).
10. Ford, H. & Alexander, J. M. Simplified hot rolling calculations. *J. Inst. Met.* **92,** 397–404 (1964).
11. Tselikov, A. *Stress and strain in metal rolling*. (Mir Publishers, 1967).
12. Wusatowski, Z. *Fundamentals of rolling*. (Pergamon Press, 1969).
13. Larke, E. *The rolling of strip, sheet and plate*. (Chapman & Hall, 1967).
14. Denton, B. K. & Crane, F. A. . Roll load and torque in the hot rolling of steel strip. *J. Iron Steel Inst.* **210,** 606–616 (1972).
15. Roberts, W. *Hot Rolling of Steel*. (CRC Press, 1977).
16. Schey, J. *Introduction to Manufacturing Processes*. (McGraw-Hill, 1987).
17. Shahani, A. R., Setayeshi, S., Nodamaie, S. A., Asadi, M. A. & Rezaie, S. Prediction of influence parameters on the hot rolling process using finite element method and neural network. *J. Mater. Process. Technol.* **209,** 1920–1935 (2009).
18. Johnson, G. R. & Cook, W. H. A constitutive model and data for metals subjected to large strains, high strain rates and high temperatures. in *7th International Symposium on Ballistics (1983)* **21,** 541–547 (1983).
19. Wang, X. *et al.* Modeling of mass flow behavior of hot rolled low alloy steel based on combined Johnson-Cook and Zerilli-Armstrong model. *J. Mater. Sci.* **52,** 2800–2815 (2017).
20. Siddhartha, R. Mechanics of Rolling. in *Principles and Applications of Metal Rolling* 30–103 (Cambridge University Press, 2015).
21. Wang, X. & Buchely, M. *Hot Working Modeling and Simulation*. (IAB Meeting, PSMRC, Missouri University of S&T, 2017).
22. Buchely, M. F., Van Aken, D. C., O'Malley, R. J., Lekakh, S. & Chandrashekhara, K. Hot Rolling Effect upon the High Temperature Johnson-Cook Strength and Failure Models for a 15V38 Grade Steel. in *Materials Science and Technology MS&T17* 1045–1053 (2017). doi:10.7449/2017/MST_2017_1045_1053